

Keshav Sundararaman

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EDUCATION

University of Texas at Arlington, M.S. Computer Science

Fall 2025 – Summer 2027

- GPA: 4.0 / 4.0 — Coursework: Data Engineering for Business Applications, Machine Learning, Pattern Recognition, Database Systems
- Machine Learning Specialization, Stanford University & DeepLearning.AI (Coursera) — Google Cloud Data Analytics Certificate (in progress)

Amrita Vishwa Vidyapeetham, B.Tech Computer Science

Oct 2020 – June 2024

- CGPA: 8.06 / 10 — Thesis: Ensemble ML Pipeline Generation using DSGE

TECHNICAL SKILLS

- **Programming:** Python, SQL, JavaScript
- **Data Analysis:** Pandas, NumPy, Matplotlib, Excel, Databricks, Apache Spark
- **Cloud:** Google Cloud Platform, BigQuery (in progress)
- **Machine Learning:** scikit-learn, PyTorch (foundational)
- **Tools:** Git, Linux

EXPERIENCE

Graduate Research Assistant, UTARI, UT Arlington

Jan 2026 – May 2026

- Built a pipeline to sync and process RGB and depth sensor streams from ROS bag files, running object detection (YOLO-World), segmentation (MobileSAM), and depth projection into persistent 3D maps for robot perception.
- Profiled the pipeline and found detection was the bottleneck at under 10 fps with GroundingDINO; replaced it with YOLO-World and brought average throughput to around 25 fps.
- Separated map building into two passes (geometry first, semantics second) and added a CLI, ROS2 Nav2 export, and multiple output formats including floor-plan images, annotated video, and JSON summaries.

Research Assistant, Sony Industry-Academia Project, India

May 2023 – Dec 2023

- Collected embedded hardware sensor data over network interfaces and processed it into structured datasets for analysis.
- Analyzed food drying behavior from the collected data and presented findings to project stakeholders.

PROJECTS

BestPick — Chrome Extension — Chrome Web Store — GitHub

Sept 2026

- Built a Chrome extension (Manifest V3, vanilla JavaScript) that scans all variants on an Amazon product page and finds the lowest price across every condition: New, Like New, Very Good, Good, and Acceptable.
- Used a pool of 5 concurrent workers with randomized delays and exponential backoff to handle rate limiting; parsed prices by matching condition heading text rather than CSS class names, which breaks when Amazon updates its markup.
- Released v1.1 within 24 hours of the initial launch to add support for size-and-color variant grids and bundle listings; the extension is live on the Chrome Web Store.

Abnormal Move Prediction in S&P 500 Stocks — GitHub

May 2026

- Built an end-to-end ML pipeline in Python to predict abnormal next-day price moves across 505 S&P 500 stocks using 619K+ rows of daily OHLCV data spanning 5 years, with 12 features engineered from price and volume history.
- Split data strictly by time to avoid leakage and tuned four model families (L1/L2 logistic regression, linear SVM, random forest) on validation F1.
- Random Forest scored best (test AUC 0.72, F1 0.31 at a 2% move threshold); feature importance and L1 sparsification showed volatility and price-range features carried the most predictive signal.